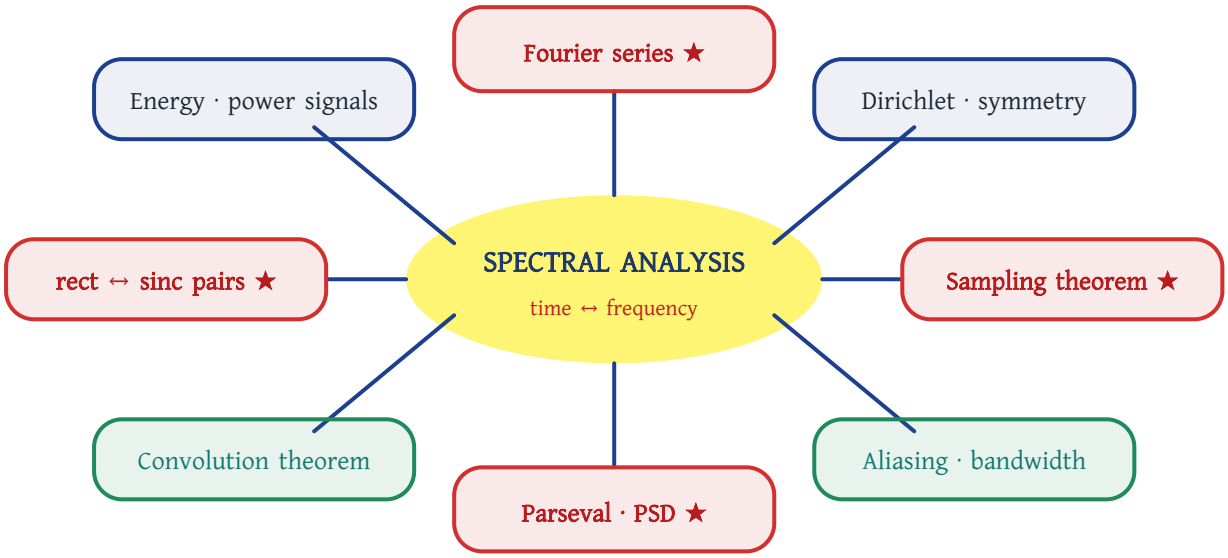
Spectral Analysis

Fourier series · Fourier transform · Convolution · Parseval · Sampling: exam-ready in 12 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

Syllabus coverage

"Spectral analysis: **Fourier series**, **sampling function**, power spectrum, **Fourier transform**, convolution, **Parseval's theorem**". These notes add signal classification, the sampling theorem with aliasing, and bandwidth. Entropy and channel capacity are in the Information Theory lesson; AM/FM, PCM and TDM/FDM are in the Modulation & Multiplexing lesson.

High-yield list

① Nyquist rate = $2f_{max}$
② rect ↔ sinc, δ ↔ 1
③ even → cosines only
④ convolution ↔ product
⑤ Parseval: E in t = E in f
⑥ alias $f_a = |f - k \cdot f_s|$

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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1 Signals & Their Classification

Definition: A **signal** x(t) is a function of time that carries information. **Spectral analysis** describes it by its **frequency content** (spectrum) instead of its waveform; Fourier analysis is the bridge.

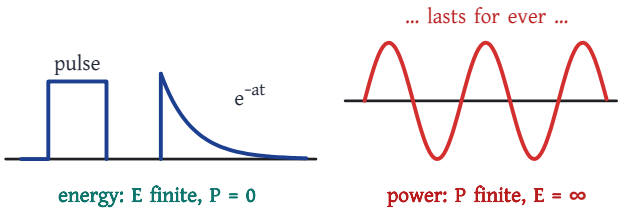
Pair	Test	Example
Continuous / discrete-time	every t / only $t = nT_s$	speech x(t) / samples x[n]
Analog / digital	any amplitude / finite levels	mic output / PCM bits
Deterministic / random	formula / statistics only	$A \cos \omega_0 t$ / noise
Periodic / aperiodic	$x(t + T_0) = x(t)$ for all t	sine wave / one pulse
Even / odd	$x(-t) = x(t)$ / $x(-t) = -x(t)$	$\cos \omega_0 t$ / $\sin \omega_0 t$

2 Energy & Power Signals

Formulas (1 Ω): $E = \int |x(t)|^2 \, dt$ (J) · $P = \lim_{T\to\infty} (1/T) \int_{-T/2}^{T/2} |x(t)|^2 \, dt$ (W)

energy signal: $0 < E < \infty$, $P = 0$ · **power signal:** $0 < P < \infty$, $E = \infty$

Signal	E	P
$A \, \mathrm{rect}(t/\tau)$	$A^2\tau$	0
$e^{-at}u(t)$, $a > 0$	$1/(2a)$	0
$A \cos(\omega_0 t + \theta)$	∞	$A^2/2$
DC level A	∞	A^2
unit step u(t)	∞	1/2
$t \cdot u(t)$, $e^{at}u(t)$	∞	∞ (neither)



Exam trap: Every periodic signal is a **power** signal, never an energy signal. A signal cannot be both; some are **neither** ($t \cdot u(t)$). Phase θ and frequency do not change the power of $A \cos(\omega_0 t + \theta)$.

Periodicity, even/odd parts & the power of a sum

Periodicity of a sum: $x_1 + x_2$ is periodic only if T_1/T_2 is **rational**; then $T_0 = \mathrm{LCM}(T_1, T_2)$. $\cos(2\pi t/3) + \sin(2\pi t/5) \rightarrow T_0 = 15$ s; $\cos t + \cos \pi t \rightarrow$ ratio π → **not periodic**.

Even/odd parts: $x_e = [x(t) + x(-t)]/2$, $x_o = [x(t) - x(-t)]/2$. even × even = even, odd × odd = even, even × odd = odd; $\int_{-a}^{a}$ (odd) $dt = 0$.

Memory trick: "**Energy signals die out, power signals live for ever.**" Finite duration or decaying → energy; repeating or constant → power.

Worked example ✍️: $x(t) = 3\cos(100\pi t) + 4\sin(200\pi t)$: $f = 50$ Hz and 100 Hz, so $T_0 = 20$ ms. Powers add (different frequencies): $P = 9/2 + 16/2 =$ **12.5 W**

3 Fourier Series: Three Equivalent Forms

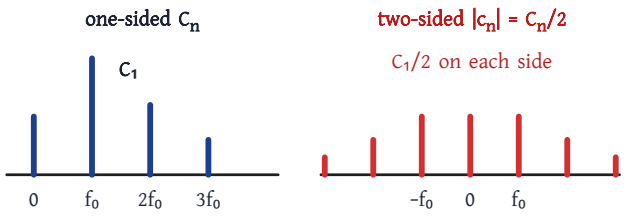
Definition: A periodic signal $x(t)$ with period T_0 (meeting Dirichlet's conditions) equals a **DC term plus harmonics** at nf_0, where $\mathbf{f_0 = 1/T_0}$ and $\omega_0 = 2\pi/T_0$. Its spectrum is a set of **discrete lines** (a line spectrum).

Form	Series	Coefficients
Trigonometric	$a_0 + \Sigma_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$	a_0, a_n, b_n (box below)
Compact (polar)	$C_0 + \Sigma\, C_n \cos(n\omega_0 t + \theta_n)$	$C_n = \sqrt{(a_n^2 + b_n^2)}$, $\theta_n = -\tan^{-1}(b_n/a_n)$
Exponential	$\Sigma_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$	$c_n = (1/T_0)\int_{T0} x(t)\, e^{-jn\omega_0 t}\, dt$

Coefficients (∫ over one period): $a_0 = \mathbf{(1/T_0)}\int x(t)\, dt$ = DC (average) · $a_n = \mathbf{(2/T_0)}\int x(t) \cos n\omega_0 t\, dt$ · $b_n = (2/T_0)\int x(t) \sin n\omega_0 t\, dt$

Links: $c_0 = a_0$ · $c_n = (a_n - jb_n)/2$ · real $x(t)$: $c_{-n} = c_n^* \Rightarrow \mathbf{|c_n| = C_n/2}$ $(n \geq 1)$

Exam trap: In the **two-sided** (exponential) spectrum each line at $\pm nf_0$ is **half** the one-sided amplitude C_n; only the DC line keeps its height. Negative frequencies are a mathematical bookkeeping device, not a physical band.



4 Dirichlet Conditions & Symmetry

Dirichlet conditions (**sufficient, not necessary**), per period:

1. absolutely integrable: $\int_{T0} |x(t)|\, dt < \infty$
2. finite number of maxima and minima
3. finite number of **finite** discontinuities

At a jump the series converges to the **mid-point** $[x(t^+) + x(t^-)]/2$.

Symmetry	What survives
Even: $x(-t) = x(t)$	a_0, a_n; $\mathbf{b_n = 0}$
Odd: $x(-t) = -x(t)$	b_n only
Half-wave: $x(t \pm T_0/2) = -x(t)$	**odd n only**
Even + half-wave	odd cosines only

Memory trick: **Even ↔ cos, Odd ↔ sin** (cos is even, sin is odd). "**Half**-wave symmetry keeps the **odd** harmonics."

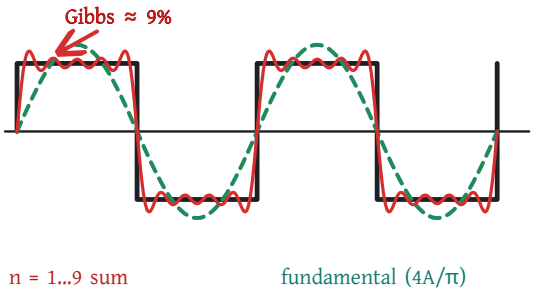
Shortcuts: Adding a DC level changes only a_0. A time shift changes only the **phases** θ_n, never $|c_n|$. Scaling time changes f_0, not the coefficients.

5 Worked Example: the Square Wave

Odd square wave, $x(t) = +A$ for $0 < t < T_0/2$ and $-A$ for $T_0/2 < t < T_0$.

1. Odd ⇒ $a_0 = a_n = 0$; half-wave symmetric ⇒ only odd n survive
2. $b_n = (2/T_0)[\int_0^{T0/2} A \sin n\omega_0 t\, dt - \int_{T0/2}^{T0} A \sin n\omega_0 t\, dt] = (2A/n\pi)(1 - \cos n\pi)$
3. n odd: $b_n = \mathbf{4A/(n\pi)}$; n even: $b_n = 0$

Result: $x(t) = \mathbf{(4A/\pi)\,[\sin \omega_0 t + (1/3)\sin 3\omega_0 t + (1/5)\sin 5\omega_0 t + \ldots]}$



- Total power $P = \mathbf{A^2}$ ($|x| = A$ always)
- Fundamental: $(4A/\pi)^2/2 = 8A^2/\pi^2 =$ **81.1%** of P
- n = 1, 3, 5 together: **93.3%**
- Amplitudes fall as **1/n** (a jump); a triangle wave (a kink) as **$1/n^2$** ($8A/(\pi^2 n^2)$); FT: rect → sinc ∝ 1/f, triangle → $\text{sinc}^2 \propto 1/f^2$. Smoother ⇒ **narrower spectrum**

Gibbs phenomenon: Near a jump a truncated series overshoots by **≈ 9% of the jump** (peak ≈ 1.18A). More terms make the ripple narrower, **never smaller**.

6 Rectangular Pulse Train & Power from the Series

Pulses of height A, width τ, period T_0; duty cycle $d = \tau/T_0$:

$\mathbf{c_n = A{\cdot}d{\cdot}\text{sinc}(n{\cdot}d)}$

- lines every $f_0 = 1/T_0$; DC = A·d
- sinc envelope with **nulls at multiples of 1/τ**
- d = 1/4 → lines n = ±4, ±8 ... are missing

Parseval for the series:

$P = (1/T_0)\int_{T0}|x|^2\, dt = \mathbf{\Sigma\, |c_n|^2}$

$= a_0^2 + \Sigma\,(a_n^2 + b_n^2)/2 = C_0^2 + \Sigma\, C_n^2/2$

each harmonic's power adds independently (orthogonality)

Worked example ✍️: $x(t) = 1 + 2\cos(2\pi\cdot 50t) + 4\sin(2\pi\cdot 150t)$: $T_0 = 20$ ms; $a_0 = 1$, $a_1 = 2$, $b_3 = 4 \rightarrow c_{\pm 1} = 1$, $c_{\pm 3} = \mp 2j$ ($|c_3| = 2$); $P = 1 + 2^2/2 + 4^2/2 =$ **11 W**.

Exam trap: Keep τ fixed and let $T_0 \rightarrow \infty$: lines crowd together (spacing $1/T_0 \rightarrow 0$) and the line spectrum becomes the **continuous** Fourier transform. That is exactly how the FT is born (below).

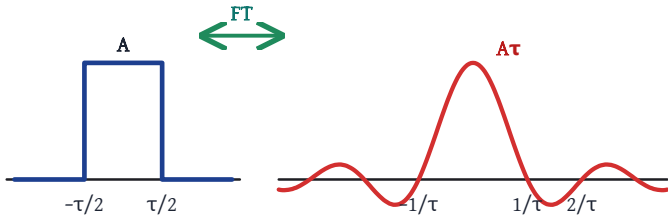
7 The Fourier Transform

Transform pair: $\mathbf{X(f) = \int x(t)\, e^{-j2\pi ft}\, dt}$ · $\mathbf{x(t) = \int X(f)\, e^{j2\pi ft}\, df}$ (limits $-\infty$ to ∞)

In ω: $X(\omega) = \int x\, e^{-j\omega t}\, dt$, $x(t) = \mathbf{(1/2\pi)} \int X(\omega)\, e^{j\omega t}\, d\omega$

- For **aperiodic** signals; X(f) is a continuous **spectral density** (V/Hz): $|X(f)|$ = amplitude spectrum, $\angle X(f)$ = phase spectrum
- Real $x(t) \Rightarrow X(-f) = X^*(f)$: **$|X|$ is even, phase is odd**
- Exists if $\int |x|\, dt < \infty$ (+ Dirichlet); every energy signal has one. Power signals (DC, cos, u(t)) need impulses $\delta(f)$

Sinc & the "sampling function": $\text{sinc}(x) = \sin(\pi x)/(\pi x)$ (normalised) · **$\text{Sa}(x) = \sin x / x$**, the **sampling function** of the syllabus. $\text{sinc}(0) = \text{Sa}(0) = 1$; sinc zeros at $x = \pm 1, \pm 2 \ldots$; Sa zeros at $\pm\pi, \pm 2\pi \ldots$; $\text{sinc}(x) = \text{Sa}(\pi x)$; $\int \text{sinc}(x)\, dx = 1$.



A rect(t/τ) ↔ Aτ sinc(fτ): nulls every 1/τ

Key transform pairs (memorise the starred ones)

x(t)	X(f)
$\delta(t)$ ★	1
1 (DC) ★	$\delta(f)$
$\delta(t - t_0)$	$e^{-j2\pi ft_0}$
$e^{j2\pi f_0 t}$	$\delta(f - f_0)$
$\cos 2\pi f_0 t$ ★	$\frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$
$\sin 2\pi f_0 t$	$(1/2j)[\delta(f - f_0) - \delta(f + f_0)]$
u(t)	$1/(j2\pi f) + \frac{1}{2}\delta(f)$
sgn(t)	$1/(j\pi f)$

x(t)	X(f)
A rect(t/τ) ★	**Aτ sinc(fτ)**
2W sinc(2Wt) ★	rect(f/2W) (band ±W)
$\Lambda(t/\tau)$ triangle	$\tau\, \text{sinc}^2(f\tau)$
$e^{-at}u(t)$, $a > 0$ ★	$1/(a + j2\pi f)$
$e^{-a\lvert t\rvert}$	$2a/(a^2 + 4\pi^2 f^2)$
$e^{-\pi t^2}$ (Gaussian)	$e^{-\pi f^2}$ (itself)
$\Sigma\, \delta(t - nT_0)$	$f_0\, \Sigma\, \delta(f - nf_0)$

$\Lambda(t/\tau)$: triangle of height 1 and half-width τ (base 2τ). rect(t/τ): height 1 for $|t| < \tau/2$.

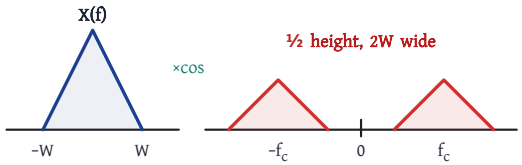
Exam trap: $\delta(t)$ (shortest signal) has a **flat spectrum containing every frequency**; DC (longest) has only $f = 0$. Books use both $\text{sinc}(x) = \sin \pi x/\pi x$ and $\sin x/x$: check which one the options use.

8 Properties of the Fourier Transform

Property	x(t)	X(f)
Linearity	$a\,x_1(t) + b\,x_2(t)$	$a\,X_1(f) + b\,X_2(f)$
Time shift ★	$x(t - t_0)$	$X(f)\,e^{-j2\pi f t_0}$ ($\|X\|$ unchanged)
Frequency shift ★	$x(t)\,e^{j2\pi f_0 t}$	$X(f - f_0)$
Modulation ★	$x(t)\cos 2\pi f_c t$	$\mathbf{\frac{1}{2}[X(f - f_c) + X(f + f_c)]}$
Time scaling ★	$x(at)$	$(1/\|a\|)\,X(f/a)$
Time reversal	$x(-t)$	$X(-f)$
Duality ★	$X(t)$	$x(-f)$
Conjugation	$x^*(t)$	$X^*(-f)$
Differentiation	dx/dt	$j2\pi f\,X(f)$
Integration	$\int_{-\infty}^{t} x(\lambda)\,d\lambda$	$X(f)/(j2\pi f) + \frac{1}{2}X(0)\delta(f)$
Convolution ★	$x_1(t) * x_2(t)$	$X_1(f) \cdot X_2(f)$
Multiplication ★	$x_1(t) \cdot x_2(t)$	$X_1(f) * X_2(f)$
Area / value at 0	$\int x(t)\,dt = X(0)$	$x(0) = \int X(f)\,df$

Memory tricks: ① "A **delay only twists the phase**; a frequency shift is modulation." ② "**Squeeze time, stretch spectrum**": $x(2t)$ has twice the bandwidth. ③ Duality: rect ↔ sinc works both ways.

Worked examples ✍️: ① Pulse from 0 to τ = $A\,\text{rect}((t - \tau/2)/\tau) \leftrightarrow \mathbf{A\tau\,\text{sinc}(f\tau)\,e^{-j\pi f\tau}}$ (same $|X|$, linear phase). ② RF pulse $A\,\text{rect}(t/\tau)\cos 2\pi f_c t \leftrightarrow (A\tau/2)[\text{sinc}((f - f_c)\tau) + \text{sinc}((f + f_c)\tau)]$. ③ If $x(t) \leftrightarrow X(f)$, then $x(3t) \leftrightarrow (1/3)X(f/3)$: bandwidth × 3. AM/DSB-SC use ②: see the Modulation lesson.



modulation: the baseband spectrum moves to $\pm f_c$

9 Bandwidth

- **Absolute:** X(f) = 0 outside B (ideal band-limited signals only)
- **3-dB (half-power):** where |X| falls to 1/√2 of peak
- **Null-to-null / first-null:** width of the main lobe
- **99% (fractional-power):** band holding 99% of the power

Signal / system	Bandwidth
pulse, width τ	first null **1/τ**
NRZ data, R bit/s	first null **R** Hz
RZ (50% duty) data	first null **2R** Hz
sinc pulses, R symbol/s	**R/2** (minimum)
RC low-pass filter	$f_{3dB} = 1/(2\pi RC)$
$x(t)\cos 2\pi f_c t$, x band W	2W around f_c

Time-bandwidth product: τ·B ≈ **constant**. A pulse of width τ has first-null bandwidth **1/τ** (baseband): 1 μs → 1 MHz, 1 ns → 1 GHz. No signal is both time-limited and band-limited.

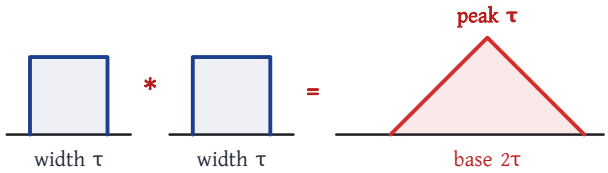
Exam trap: Baseband bandwidth counts **positive** frequencies only: nulls at ±1/τ give B = 1/τ, not 2/τ. After modulation the band becomes **2/τ** around f_c.

RC: R = 1 kΩ, C = 0.1 μF → f_{3dB} ≈ 1.59 kHz.

10 Convolution

Definition: **$y(t) = x(t) * h(t) = \int x(\tau)\, h(t-\tau)\, d\tau$**. For a **linear time-invariant (LTI)** system with impulse response h(t), the output is the input convolved with h(t). Discrete: $y[n] = \Sigma_k\, x[k]\, h[n-k]$.

1. **Flip** h(τ) → h(-τ)
2. **Shift** by t → h(t - τ)
3. **Multiply** by x(τ)
4. **Integrate** (area) → y(t); repeat for every t



rect * rect = triangle (widths add)

- Commutative, associative, distributive
- $x * \delta(t) = x(t)$ (identity) · $x * \delta(t - t_0) = x(t - t_0)$
- Width of y = width of x + width of h
- Area of y = area of x × area of h
- Discrete length: **N + M − 1**

Convolution theorem:
$x * h \leftrightarrow X \cdot H$ and $x \cdot h \leftrightarrow X * H$
⇒ H(f) = Y(f)/X(f) = **transfer function** = FT of h(t). Filters **multiply** spectra; sampling and windowing (products in time) **convolve** spectra.

Worked example 1 ✍️: [1, 2, 3] * [1, 1]:
y[0] = 1 · y[1] = 2 + 1 = 3 · y[2] = 3 + 2 = 5 · y[3] = 3
y = **[1, 3, 5, 3]**, length 3 + 2 − 1 = 4 ✓

Worked example 2 ✍️: $u(t) * e^{-at}u(t) = \int_0^t e^{-a\tau}\, d\tau$ = **$(1/a)(1 - e^{-at})\, u(t)$**: the step response of an RC low-pass (a = 1/RC).

Exam trap: Squaring a signal doubles its bandwidth: $x^2 \leftrightarrow X * X$, so a 3 kHz signal squared occupies 6 kHz. A product $x_1 x_2$ occupies $W_1 + W_2$.

Correlation: convolution without the flip

Definition: $R_{xy}(\tau) = \int x(t)\, y^*(t - \tau)\, dt = x(\tau) * y^*(-\tau)$: **no flip**, **not commutative** ($R_{xy}(\tau) = R_{yx}^*(-\tau)$), $\mathcal{F}\{R_{xy}\} = X \cdot Y^*$. **Autocorrelation** $R_x(\tau)$: even for real x, maximum at $\tau = 0$, **R(0) = energy**, $\mathcal{F}\{R_x\} = |X(f)|^2$. IT link: radar, GPS and CDMA receivers find a delay at the **correlation peak** (matched filter).

11 Parseval's Theorem

Statement: The energy (or power) of a signal is the **same whether summed over time or over frequency**. Each frequency component contributes independently, because the complex exponentials are orthogonal.

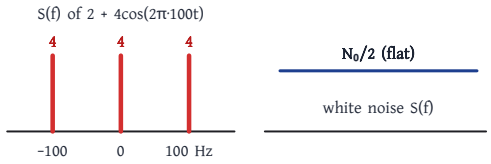
Case	Time domain	Frequency domain
FT, energy (Rayleigh) ★	$E = \int \lvert x(t)\rvert^2\, dt$	$= \int \lvert X(f)\rvert^2\, df$
Fourier series, power ★	$P = (1/T_0) \int_{T0} \lvert x(t)\rvert^2\, dt$	$= \Sigma \lvert c_n\rvert^2 = C_0^{\,2} + \Sigma\, C_n^{\,2}/2$
DFT	$\Sigma \lvert x[n]\rvert^2$	$= (1/N)\, \Sigma \lvert X[k]\rvert^2$

Worked examples ✍️:

1. $x(t) = 2 + 4\cos(2\pi \cdot 100t)$: $c_0 = 2$, $c_{\pm 1} = 2 \rightarrow P = 2^2 + 2^2 + 2^2 =$ **12 W** ($= 2^2 + 4^2/2$ ✓)
2. $\int \text{sinc}^2(t)\, dt = \int \text{rect}^2(f)\, df =$ **1**: integrate in whichever domain is easier
3. $e^{-at}u(t)$: $E = \int_0^\infty e^{-2at}\, dt = 1/(2a) = \int df/(a^2 + 4\pi^2 f^2)$ ✓; $a = 2 \rightarrow$ **0.25 J**. A rect pulse keeps **≈ 90%** of its energy in the main lobe $|f| < 1/\tau$

12 Energy & Power Spectral Density

	ESD (energy signals)	PSD (power signals)
Definition	$\Psi(f) = \lvert X(f)\rvert^2$ (J/Hz)	$S(f) = \lim_{T\to\infty} \lvert X_T(f)\rvert^2/T$ (W/Hz)
Autocorrelation	$R(\tau) = \int x(t)\, x^*(t - \tau)\, dt$; $R(0) = E$	time-averaged $R(\tau)$; **R(0) = P**
Wiener–Khinchin	$\Psi(f) = \mathcal{F}\{R(\tau)\}$	**$S(f) = \mathcal{F}\{R(\tau)\}$**
Through H(f)	$\Psi_y = \lvert H(f)\rvert^2\, \Psi_x$	$S_y = \lvert H(f)\rvert^2\, S_x$



periodic signal = PSD lines; white noise = flat PSD

Periodic: $S(f) = \Sigma\ |c_n|^2\ \delta(f - nf_0)$ (lines). **White noise:** $S(f) = N_0/2$, $R(\tau) = (N_0/2)\delta(\tau)$; after an ideal LPF of bandwidth B the noise power is $\mathbf{N_0B}$ (the N in Shannon's $C = B\ \log_2(1 + S/N)$).

Exam trap: $|X(f)|^2$ is the **energy** spectral density; for a power signal divide by T and take the limit. PSD is real, even, ≥ 0: **phase is lost**. Remember: "**R and S are a Fourier pair; R(0) is the total.**"

Worked examples ✍️:

1. A $\cos(2\pi f_0 t + \Theta)$: $R(\tau) = (A^2/2)\cos 2\pi f_0\tau \rightarrow S(f) = (A^2/4)[\delta(f - f_0) + \delta(f + f_0)]$; $P = R(0) = \mathbf{A^2/2}$ (Θ is lost)
2. White noise $N_0/2$ through an RC low-pass: $S_y = (N_0/2)/(1 + (2\pi fRC)^2) \rightarrow P_y = \mathbf{N_0/(4RC)}$; noise-equivalent bandwidth $B_N = 1/(4RC) = (\pi/2)\ f_{3dB}$

13 The Sampling Theorem

Statement: A signal **band-limited to W Hz** ($X(f) = 0$ for $|f| > W$) is fully recovered (ideal LPF) from samples taken at $\mathbf{f_s \geq 2W}$. **Nyquist rate = 2W** (minimum f_s); **Nyquist interval = 1/(2W)** (maximum T_s). Nyquist 1928, Shannon 1949.

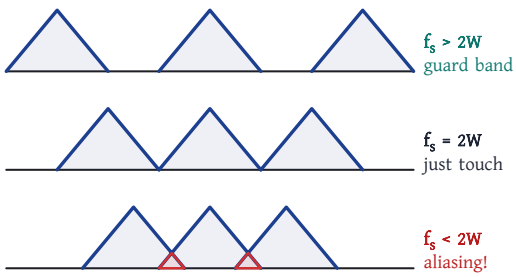
Ideal (impulse) sampling: $x_s(t) = x(t) \cdot \Sigma\ \delta(t - nT_s)$

⇒ $\mathbf{X_s(f) = f_s\ \Sigma\ X(f - n\ f_s)}$: copies of the spectrum at every multiple of f_s (a product in time is a convolution in frequency).

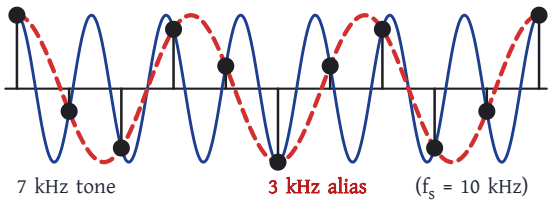
Recovery: LPF cut-off $W \leq f_c \leq f_s - W$; guard band = $f_s - 2W$. Ideal interpolation: $x(t) = \Sigma\ x(nT_s)\ \mathrm{sinc}((t - nT_s)/T_s)$.

Aliasing: if $f_s < 2W$ the copies overlap; a tone f appears at $\mathbf{f_a = |f - k{\cdot}f_s|}$ (k = nearest integer, $f_a \leq f_s/2$). Cure: **anti-aliasing LPF** (cut-off $f_s/2$) **before** the sampler + sample above 2W.

Practical: natural (gated) sampling keeps the baseband copy undistorted; **flat-top** (sample-and-hold, PAM) multiplies it by $\mathrm{sinc}(f\tau)$ → **aperture effect**, fixed by an equaliser.



spectral copies every f_s: gap, touch, overlap



same samples, two sinusoids: the LPF rebuilds 3 kHz

Nyquist-rate & alias problems (high-yield MCQs)

Signal	Working	Answer
band-limited to 4 kHz	2 × 4 kHz	**8 kHz**
$\cos(200\pi t) + \cos(600\pi t)$	100 Hz, 300 Hz → f_{max} = 300 Hz	**600 Hz**
speech, 3.4 kHz	T_s = 1/6800 ≈ 147 μs; telephony samples at 8 kHz	**6.8 kHz**
CD audio, 20 kHz	uses 44.1 kHz (guard band 4.1 kHz)	**40 kHz**
$\cos^2(2\pi\cdot1000t)$	= ½ + ½cos(2π·2000t) → f_{max} = 2 kHz	**4 kHz**
x_1(3 kHz) · x_2(5 kHz)	product: W = 3 + 5 = 8 kHz (a sum: 5 kHz)	**16 kHz**
sinc(200t); sinc^2(200t)	W = 100 Hz; squaring doubles W to 200 Hz	**200; 400 Hz**
60 Hz tone, f_s = 100 Hz	|60 − 100| = 40 Hz	alias **40 Hz**

Exam trap: A pure tone sampled at **exactly** 2f can give all-zero samples (sin nπ = 0), so use $f_s > 2f$ strictly. Chain: anti-aliasing LPF → sampler → quantizer → encoder (PCM details: Modulation lesson).

14 DFT & FFT (the digital spectrum)

DFT: $X[k] = \Sigma_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}$, k = 0 … N − 1 · inverse has the factor **1/N** · bin k ↔ f = k·f_s/N · resolution **Δf = f_s/N**

- Radix-2 FFT (Cooley–Tukey, **1965**, N a power of 2): **$(N/2)\log_2 N$** complex multiplications instead of **N^2**
- N = 1024: 1,048,576 → 5,120, **≈ 205× faster**
- f_s = 8 kHz, N = 256 → Δf = **31.25 Hz**
- DFT product ↔ **circular** convolution
- IT link: OFDM (Wi-Fi, 4G/5G) uses IFFT/FFT

Worked example ✍️: 4-point DFT of x = [1, 1, 0, 0]: $X[k] = 1 + e^{-j\pi k/2}$ → X = **[2, 1 − j, 0, 1 + j]**. Parseval check: $\Sigma|x|^2 = 2 = (1/4)(4 + 2 + 0 + 2)$ ✓. X[0] = Σ x[n] is always the DC sum.

Night-Before-the-Exam Sheet

Formula sheet:

$E = \int |x|^2\,dt \cdot P = \lim (1/T)\int |x|^2\,dt$

$A\cos(\omega_0 t + \theta) \to P = A^2/2 \cdot DC\ A \to A^2$

$f_0 = 1/T_0 \cdot \omega_0 = 2\pi/T_0$

$a_0 = (1/T_0)\int x \cdot a_n,\ b_n: (2/T_0)\int \ldots$

$c_n = (a_n - jb_n)/2 \cdot |c_n| = C_n/2$

Square wave: $b_n = 4A/(n\pi)$, odd n

$X(f) = \int x\, e^{-j2\pi ft}\,dt$

$x(t) = \int X\, e^{j2\pi ft}\,df$

$A\,\mathrm{rect}(t/\tau) \leftrightarrow A\tau\,\mathrm{sinc}(f\tau)$

$\delta(t) \leftrightarrow 1 \cdot 1 \leftrightarrow \delta(f)$

$x(t - t_0) \leftrightarrow X\, e^{-j2\pi ft_0}$

$x(at) \leftrightarrow (1/|a|)X(f/a)$

$x \cos 2\pi f_c t \leftrightarrow \tfrac{1}{2}[X(f-f_c) + X(f+f_c)]$

$x * h \leftrightarrow XH \cdot xh \leftrightarrow X * H$

$E = \int |x|^2\,dt = \int |X|^2\,df$

$P = \Sigma |c_n|^2 \cdot S(f) = F\{R(\tau)\}$

$f_s \geq 2W \cdot T_s \leq 1/(2W)$

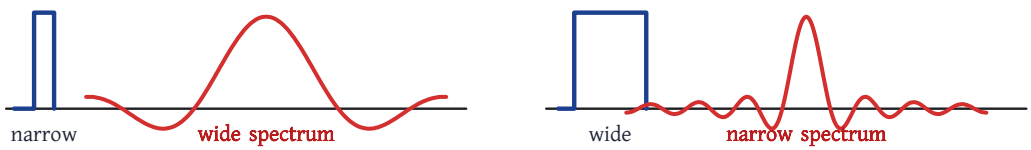
$f_a = |f - k\,f_s| \cdot \Delta f = f_s/N$

Top 12 one-liners for MCQs

1. Even signal → only DC + cosines
2. Odd signal → only sines
3. Half-wave symmetry → odd harmonics
4. Square wave harmonics ∝ 1/n
5. Gibbs overshoot ≈ 9%, never vanishes
6. FT of rect = sinc; of δ(t) = 1
7. Convolution in t = multiplication in f
8. Parseval: energy same in t and f
9. PSD = FT of autocorrelation (Wiener–Khinchin)
10. Nyquist rate = $2f_{max}$; 4 kHz → 8 kHz
11. Anti-aliasing filter goes before the sampler
12. Periodic signal = power signal

Top traps: two-sided lines are half height · $|X(f)|^2$ is ESD, not PSD · $\cos^2$ and products widen bandwidth · Nyquist **rate** (2W) ≠ Nyquist **interval** (1/2W) · tone at exactly 2f: use $f_s > 2f$ · zero-padding does not improve resolution · two sinc conventions

All tricks together: Energy signals die out, power signals live for ever · Even ↔ cos, Odd ↔ sin · half-wave keeps odd harmonics · a delay only twists the phase · squeeze time, stretch spectrum · R and S are a Fourier pair, R(0) is the total



time-bandwidth product: τ·B ≈ constant

✓ Practice MCQs (exam pattern, 10 marks)

1. Nyquist rate of $x(t) = 5\cos(1000\pi t) + 3\sin(3000\pi t)$:
(a) 1500 Hz (b) 3000 Hz
(c) 1000 Hz (d) 6000 Hz

2. The Fourier transform of $\delta(t)$ is:
(a) $\delta(f)$ (b) 0
(c) 1 (d) sinc(f)

3. The Fourier series of an odd periodic function has:
(a) only cosine terms (b) only a DC term
(c) DC and cosines (d) only sine terms

4. Convolution in time corresponds in frequency to:
(a) multiplication (b) convolution
(c) addition (d) differentiation

5. Average power of $3\cos(100\pi t) + 4\sin(200\pi t)$:
(a) 25 W (b) 12.5 W
(c) 7 W (d) 3.5 W

6. If $x(t) \leftrightarrow X(f)$, then $x(t - t_0) \leftrightarrow$
(a) $X(f - t_0)$ (b) $X(f)\, e^{j2\pi f t_0}$
(c) $X(f)\, e^{-j2\pi f t_0}$ (d) $X(f)/t_0$

7. A symmetrical square wave contains:
(a) all harmonics (b) only even harmonics
(c) odd harmonics $\propto 1/n^2$ (d) odd harmonics $\propto 1/n$

8. A 60 Hz tone sampled at 100 Hz is reconstructed as:
(a) 40 Hz (b) 60 Hz
(c) 160 Hz (d) 100 Hz

9. Energy of the pulse $A\,\mathrm{rect}(t/\tau)$:
(a) $A\tau$ (b) $A^2\tau$
(c) $A^2/2$ (d) $A^2\tau^2$

10. By Wiener–Khinchin, the PSD is the FT of the:
(a) signal itself (b) impulse response
(c) phase spectrum (d) autocorrelation

Answer key → 1-b · 2-c · 3-d · 4-a · 5-b · 6-c · 7-d · 8-a · 9-b · 10-d

Written question: Find the Fourier transform of the pulse $x(t) = A$ for $|t| < \tau/2$ (0 elsewhere) and sketch its amplitude spectrum. Using Parseval's theorem, find its energy and first-null bandwidth for $A = 2$ V, $\tau = 1$ ms. 8 marks

1. $X(f) = \int_{-\tau/2}^{\tau/2} A\, e^{-j2\pi f t}\, dt = A\cdot[e^{j\pi f\tau} - e^{-j\pi f\tau}]/(j2\pi f) = A\,\sin(\pi f\tau)/(\pi f) = \mathbf{A\tau\ \mathrm{sinc}(f\tau)}$ 3
2. Sketch: real, even; peak $A\tau$ at $f = 0$, nulls at $\pm 1/\tau, \pm 2/\tau$ ..., side lobes fall as $1/f$ 1
3. Parseval: $E = A^2\tau = \int A^2\tau^2\,\mathrm{sinc}^2(f\tau)\,df$ ($\int \mathrm{sinc}^2(f\tau)\,df = 1/\tau$) 2; $A = 2$ V, $\tau = 1$ ms: $E =$ **4 mJ**, first null $1/\tau =$ **1 kHz** ($\approx$ 90% of E inside) 2

Before the exam ✓

- ✓ Can classify energy / power signals
- ✓ Can write a_0, a_n, b_n, c_n and use symmetry
- ✓ Know 8 FT pairs and the shift, scaling, modulation rules
- ✓ Can convolve two short sequences
- ✓ Can find a Nyquist rate and an alias frequency
- ✓ Also practise: triangle-wave series, proof of Parseval, rect * rect